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```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv('Iris.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
```

```
# Column Non-Null Count Dtype
```

```
-----
```

```
0 sepal.length 150 non-null float64
```

```
1 sepal.width 150 non-null float64
```

```
2 petal.length 150 non-null float64
```

```
3 petal.width 150 non-null float64
```

```
4 variety 150 non-null object
```

```
dtypes: float64(4), object(1)
```

```
memory usage: 6.0+ KB
```

```
df.variety.value_counts()
```

```
variety
```

```
Setosa 50
```

```
Versicolor 50
```

```
Virginica 50
```

```
Name: count, dtype: int64
```

```
df.head()
```

```
sepal.length sepal.width petal.length petal.width variety 0 5.1 3.5 1.4 0.2 Setosa 1
4.9 3.0 1.4 0.2 Setosa 2 4.7 3.2 1.3 0.2 Setosa 3 4.6 3.1 1.5 0.2 Setosa 4 5.0 3.6 1.4
0.2 Setosa
```

```
features=df.iloc[:, :-1].values
```

```
label=df.iloc[:, 4].values
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2 ,random_state=42)
```

```
model_KNN=KNeighborsClassifier(n_neighbors=5)
```

```
model_KNN.fit(xtrain,ytrain)
```

```
KNeighborsClassifier()
```

```
print(model_KNN.score(xtrain,ytrain))
```

```
print(model_KNN.score(xtest,ytest))
```

```
0.9666666666666667
```

```
1.0
```

```
from sklearn.metrics import confusion_matrix
```

```
confusion_matrix(label,model_KNN.predict(features))
```

```
array([[50, 0, 0],
```

```
 [ 0, 47, 3],
```

```
 [ 0, 1, 49]])
```